EMBER SAFE | SMARTLOOP RANGE

SmartLoop-4

Operation Manual

OPERATION MANUAL



ANALOGUE ADDRESSABLE **4 loops, 512 devices - and the exact device in alarm on the display.**

INFUSIVE INDOTECH PRIVATE LIMITED

Ember Safe | Stay Alert

Document ES-SL4-OM Revision 1.0 November 2026

Important Notices

About this manual

This manual explains how to operate the Ember Safe SmartLoop-4 analogue addressable fire alarm control panel: what the lights, keys and display show, what to do in fire and fault, and the settings available to the user.

It is written for Station Masters, ESM and maintenance staff. Installation, commissioning, maintenance and troubleshooting are in the Installation, Commissioning & Maintenance Manual (ES-SL4-IM). Technical data is in the datasheets (ES-SL4-DS1, ES-SL4-DS2).

The system is designed to RDSO/SPN/217/2025 Version 3.1.

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Disclaimer

The information in this manual is believed to be accurate at the time of publication. Infusive Indotech Private Limited reserves the right to change product features and this document without notice as part of continuous product improvement. Illustrations are representative; the appearance of your product may differ. A fire alarm system is a life-safety system: its design, installation, commissioning and maintenance must be carried out by trained, competent persons in line with the fire codes and the requirements of the local authority having jurisdiction.

Symbols used in this manual

Symbol	Meaning
WARNING	Risk to life or of serious injury, or a fire alarm that may not work. Always follow.
CAUTION	Risk of damage to the equipment or of false alarms.
NOTE	Useful information for correct operation.
GOOD PRACTICE	Recommendation from Ember Safe for the best results.

Revision history

Revision	Date	Description
1.0	November 2026	First edition.

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1 Safety Information

Read this manual before operating the panel. Keep it near the panel, in the Station Master's room.

General safety

! **WARNING** The SmartLoop-4 is part of a life-safety system. Only trained and authorised railway staff may operate it at access level 2 (key switch) or above.

Never disable a loop or device, silence the alarm or switch off the power except for authorised testing, maintenance or a confirmed false alarm. Restore normal working at once. A disabled loop or device does **not** report fire.

! **WARNING** EVACUATE starts a real fire alarm on all hooters and strobes, and may start the fire suppression system if one is connected. Inform the staff before any test.

! **CAUTION** Do not open the panel: mains voltage is present inside. Opening, wiring and card replacement are for trained maintenance staff only.

Keep the panel key with the person in charge. Do not leave it in the key switch.

Disposal

Do not dispose of the panel, devices or batteries with ordinary waste. Hand them over to an authorised e-waste and battery recycler in line with the E-Waste (Management) Rules and the Battery Waste Management Rules.

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2 Product Overview

Introduction

The **Ember Safe SmartLoop-4** is an **analogue addressable** fire alarm control panel with **4 loops**. Each loop is a two-wire circuit carrying up to **128 addressable devices** - heat & smoke multi-sensors, manual call points, input modules and relay modules - so one panel watches up to **512 devices**.

Every device has its own address and reports its own analogue value. In fire the panel shows **the loop, the address and the location** of the device, for example **LOOP 2 ADDR 017 RELAY ROOM - RACK 3 LEFT**, so staff go straight to the seat of the fire.

Key features

Exact location
Loop, address, device type and a 32-character location text for every device.

512 devices
4 loops x 128 analogue addressable devices.

Isolators in every device
A short circuit on a loop takes out only the short section.

RDSO action levels
Alert, Station Master room alarm, all alarms, and suppression release with a delay.

Two-detector coincidence
Adjacent detectors are paired: one = alert, both = fire.

Event log
At least 512 fire events and 512 other events with date and time.

Always checking
Loops, devices, mains, battery, charger, earth fault and the panel itself.

Communication
TCP/IP to Zonal HQ, PC software, optional SMS alarms on GSM.

System parts

Part	What it does
SmartLoop-4 control panel	Watches the 4 loops, shows alarms and faults, sounds the hooters, operates the outputs and keeps the event log.
ES-MSD100 multi-sensor	Detects smoke and heat. Green blink = healthy, red = alarm.
ES-MCP100 manual call point	Push to raise the alarm by hand. Reset with its key.
ES-HS100 hooter cum strobe	Sounds and flashes in fire.
ES-IM100 input module	Brings LHS / LHD, aspirating, flame and probe heat detectors onto the loop.
ES-RM100 relay module	Starts fire suppression, switches off room power, signals the data logger.
ES-GSM100 GSM module (optional)	Sends SMS alarms to the mobile numbers set in the panel.
SmartLoop PC software	Site layout view, event download and configuration from a Windows PC.

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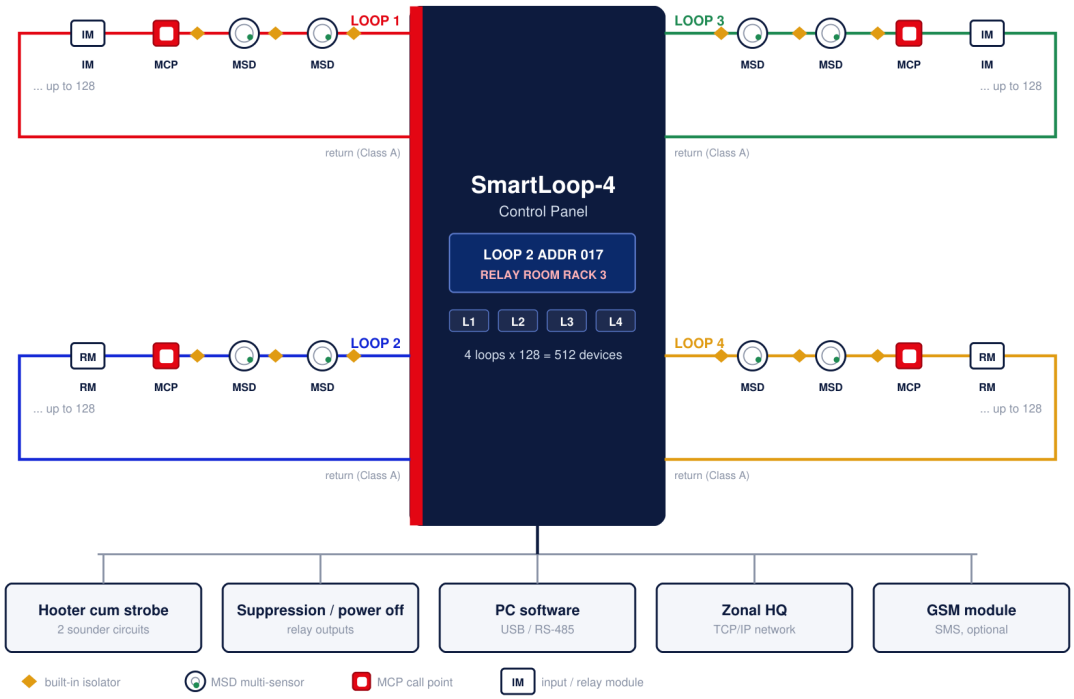


Figure 1 SmartLoop-4 system

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3 The Front Panel

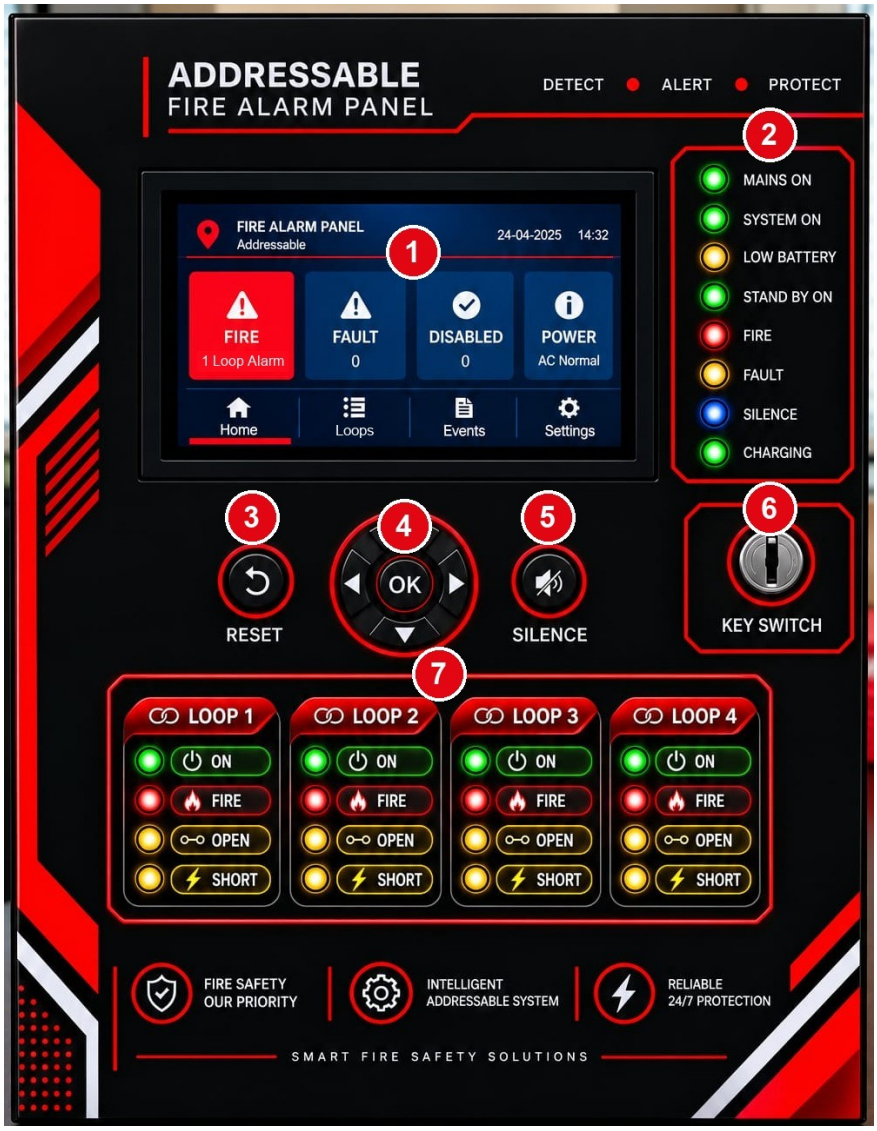


Figure 2 Front panel

No.	Part	Use
1	Colour touch display	Status, alarms, faults, event log and menus. Touch or use the arrow keys.
2	Status LEDs	Power and system state - see the table below.
3	RESET	Clears alarms and faults and restarts the loops (access level 2).
4	OK and arrow keys	Move through the screens and menus; OK selects or saves.
5	SILENCE	Silences the buzzer; at access level 2 also the hooters. Press again to sound them again.
6	Key switch	Turn to ON for access level 2 (control functions). Remove the key afterwards.
7	Loop LEDs	Four LEDs for each loop: ON, FIRE, OPEN, SHORT.

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Status LEDs

Table 1 Status LEDs

LED	Colour	On	Flashing
MAINS ON	Green	Mains supply present	-
SYSTEM ON	Green	Panel working normally	Panel fault (CPU or internal communication)
LOW BATTERY	Yellow	Battery below 21.6 V	Battery missing or not charging
STAND BY ON	Green	Running on battery (mains failed)	-
FIRE	Red	Fire confirmed (level 3 or 4)	Alert (level 1 or 2)
FAULT	Yellow	One or more faults	-
SILENCE	Blue	Hooters / buzzer silenced	-
CHARGING	Green	Battery charging	Charger fault

Loop LEDs

Table 2 Loop LEDs

LED	Colour	Meaning
ON	Green	Loop switched on and powered. Off = loop disabled.
FIRE	Red	A device on this loop is in fire.
OPEN	Yellow	Open circuit on the loop wiring. In Class A wiring the loop keeps working from both ends.
SHORT	Yellow	Short circuit on the loop. The isolators on each side cut out the short section.

Access levels

Table 3 Access levels

Level	How	Allowed
1	Anyone	See status, alarms, faults and the event log; silence the panel buzzer; lamp test.
2	Key switch ON	Level 1 + silence / re-sound hooters, RESET, EVACUATE, disable / enable loops and devices, walk test.
3	Password	Level 2 + settings: devices, location texts, coincidence pairs, action levels, outputs, delays, date and time, network, GSM numbers.
4	Manufacturer	Firmware update and factory functions.

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4 The Display

In normal working the display shows the **Home** screen: the panel name, date and time, and four tiles - **FIRE**, **FAULT**, **DISABLED** and **POWER** - with their counts. The four tabs at the bottom open the **Home**, **Loops**, **Events** and **Settings** screens.

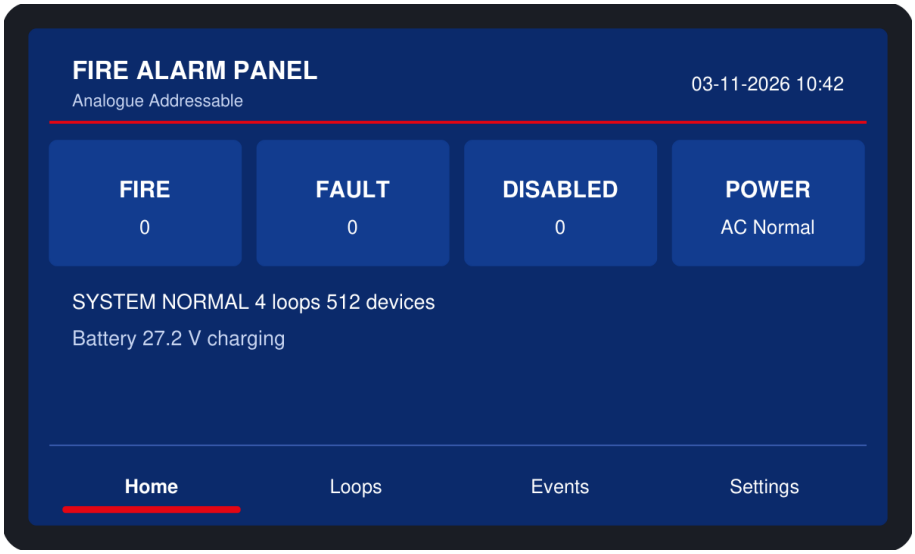


Figure 3 Home screen - normal

Tab	Shows
Home	The four status tiles, or the alarm / fault in progress. Buttons: EVACUATE, LAMP TEST, MUTE BUZZER.
Loops	Each loop with its devices: address, type, location text, state and analogue value (smoke level, temperature). Disable / enable at level 2.
Events	The event log, newest first: fire, pre-alarm, fault, disable, reset and power events with date and time. Filter by type or loop.
Settings	Menus for access level 3 (password) - see chapter 7.

Display priority

The display always shows the most important state first:

- **Fire** - every device in fire, shown one after another.
- **Pre-alarm and alert** (action levels 1 and 2).
- **Faults** - shown one after another, every 2.5 seconds.
- **Disabled** loops and devices.
- **Normal** - Home screen with date and time.

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5 Fire Alarm

What the panel does

When a detector or call point goes into fire, the panel acts on the action level set for it within 10 seconds:

Level	Started by	Panel action
Level 1	One detector of a coincidence pair, or aspirating detector stage 1	Panel buzzer and display; FIRE LED flashing
Level 2	Aspirating detector stage 2	Level 1 + hooter cum strobe in the Station Master room
Level 3	Manual call point, or aspirating detector stage 3	All hooters and strobes; power shut-down outputs (if programmed)
Level 4	Both detectors of a coincidence pair, aspirating detector stage 4, LHS / LHD module	All hooters and strobes within 10 s; suppression release after the set delay (0 to 10 min)

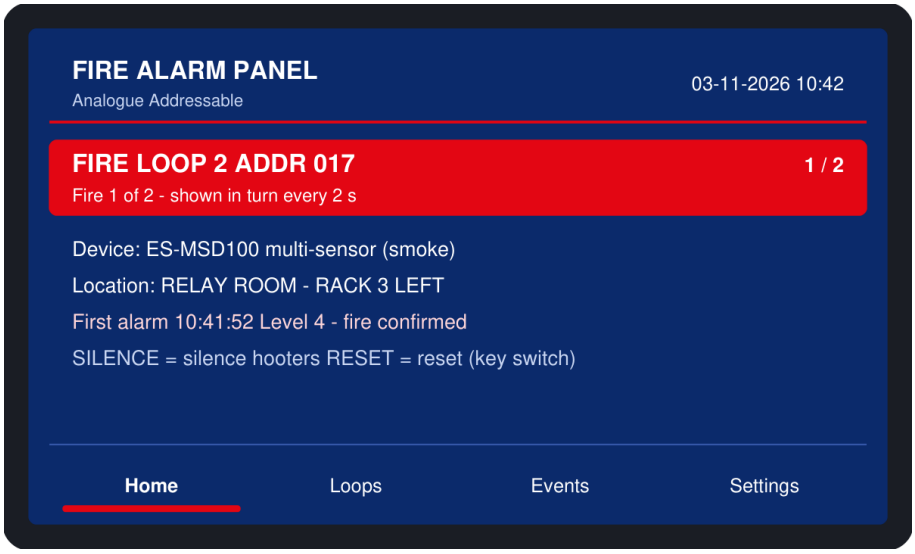


Figure 4 Fire screen - loop, address, device and location

The display shows **FIRE, the loop and the address**, the device type and its location text. If there is more than one fire, the panel shows them **one after another every 2 seconds** with the count, for example **1 / 2**. The red LED on the device in fire stays on until the panel is reset.

What to do in fire

1	Read the loop, address and location on the display. Use the arrow keys to see every fire if there is more than one.
2	Send staff to the location to confirm the fire. The red LED on the device shows which one is in alarm.
3	**Real fire:** follow the station fire orders - call the fire brigade, use the CO2 extinguishers, switch off the power to the affected equipment if required.
4	To stop the hooters: turn the **key switch** to ON and press **SILENCE**. The SILENCE LED lights. Press SILENCE again to sound the hooters again.
5	When the cause is removed (smoke cleared, call point reset with its key), press **RESET**. The panel clears the alarm and restarts the loops. If the cause is still there, the alarm comes back.

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6 Turn the key switch back to OFF and remove the key. Record the event in the station fire register.

! **WARNING** Silencing stops the hooters but **does not** clear the fire. A new fire on another device sounds the hooters again.

i **NOTE** If a suppression release delay is running (action level 4), the remaining time is shown on the display. Releasing or holding the suppression is set up by the installer as per the station fire orders.

Manual alarm - EVACUATE

To raise the alarm from the panel: turn the key switch to ON, press **EVACUATE** on the Home screen and confirm with **OK**. All hooters and strobes sound. Press RESET to stop.

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6 Faults

A fault does not mean fire, but part of the system may not be able to report fire. The FAULT LED lights, the buzzer sounds and the FAULT relay operates. Faults are shown one after another.

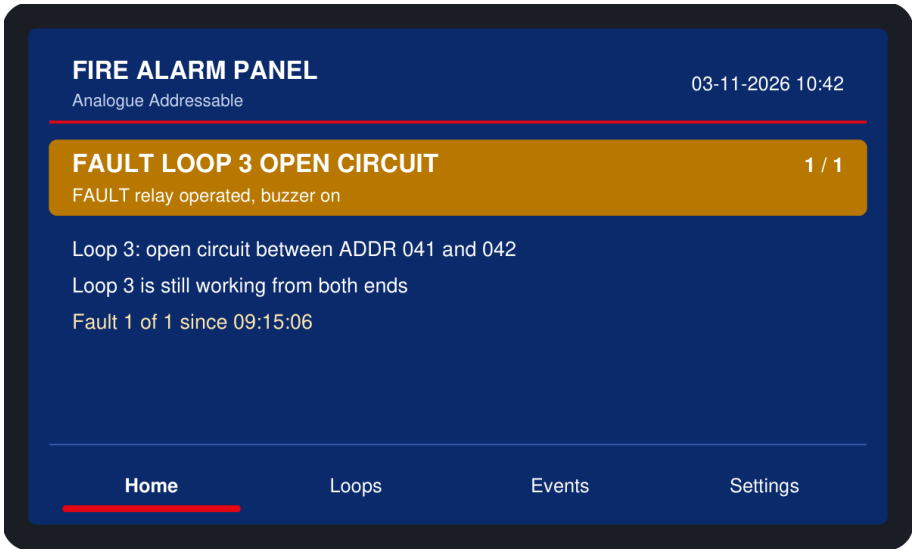


Figure 5 Fault screen

Table 4 Fault messages

Display shows	Meaning	What to do
LOOP n OPEN CIRCUIT	Loop wire broken between two addresses.	Call maintenance. In Class A the loop keeps working.
LOOP n SHORT CIRCUIT	Short on the loop; isolators have cut out the section.	Call maintenance.
DEVICE MISSING L n ADDR x	Device removed, damaged or its wiring broken.	Check the device; call maintenance.
DIRTY CHAMBER L n ADDR x	Detector needs cleaning (dust compensation at its limit).	Arrange cleaning.
EARTH FAULT	A wire touches earth.	Call maintenance.
MAINS FAULT	No mains supply; the panel runs on battery (STAND BY ON).	Restore mains.
BATTERY LOW	Battery below 21.6 V.	Restore mains; if it stays, replace the battery.
BATTERY MISSING	Battery disconnected or fuse open.	Call maintenance.
CHARGER FAULT	Battery not being charged.	Call maintenance.
SOUNDER CIRCUIT n FAULT	Hooter wiring open or short.	Call maintenance.
SYSTEM FAULT	Panel internal fault. Loop cards keep sounding alarms (degrade mode).	Call maintenance at once.

Press **SILENCE** to stop the buzzer. When the fault is cleared it goes off by itself; press **RESET** to clear latched faults.

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7 Daily Use and Settings

Lamp test

Press **LAMP TEST** on the Home screen: every LED and the buzzer come on for a few seconds. Do this daily.

Disable and enable

At level 2, open **Loops**, select a loop or a device and choose **Disable** or **Enable**. The DISABLED tile counts what is disabled; the loop ON LED goes off for a disabled loop. A disabled loop or device does not report fire - enable it again as soon as the work is finished.

Walk test

At level 2, **Settings > Walk test** lets one person test the detectors: each device tested lights its red LED and is logged, without sounding the hooters. Leave walk test when finished; the panel leaves it by itself after a set time.

Settings menu (level 3)

Table 5 **Settings menu**

Menu	What it sets
Loops and devices	Switch loops on / off, auto-learn devices, edit address, type and location text, disable / enable.
Coincidence pairs	Pairs of adjacent detectors for two-detector confirmation.
Action levels	Level of each device and input, and the outputs each level operates.
Outputs and delays	Relays, sounder circuits, suppression release delay (0 to 10 min in 0.5 min steps).
Sensitivity	Smoke sensitivity of each multi-sensor: High, Normal, Low, Very low.
Date and time	Clock; automatic time from the network when connected.
Network	TCP/IP address for the Zonal / Divisional HQ link; RS-485 sub-panels.
GSM	Up to 10 mobile numbers for SMS alarms (at least 5 required).
Display	Display and LED brightness, buzzer pattern.
Passwords	Change level 3 passwords.
Factory reset	Restore default settings (asks for confirmation).

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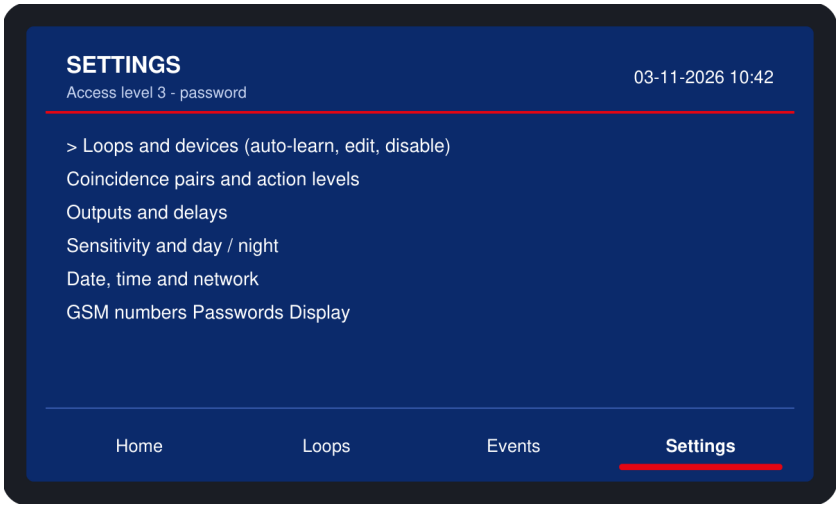


Figure 6 Settings menu

Event log and PC

The panel keeps at least **512 fire events and 512 other events** with date and time, also without power. View them under **Events**. To keep a copy, connect a PC with the SmartLoop PC software (USB, RS-485 or network) and choose **Download events**; the software shows the site layout, the events and the device values.

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8 Routine Checks by the User

When	Check
Daily	Display shows SYSTEM NORMAL, MAINS ON and SYSTEM ON lit, no FAULT. Lamp test. Faults written in the register and reported.
Weekly	Operate one manual call point (a different one each week) and check the alarm; reset. Inform staff first.
Monthly	Check the hooters and strobes sound in every room; check SMS reaches the mobile numbers (if GSM is fitted).
Every visit	No detector covered, painted or blocked; call points visible and lit; panel key with the person in charge.

Quarterly, half-yearly and yearly maintenance is done by trained maintenance staff - see the Installation, Commissioning & Maintenance Manual (ES-SL4-IM).

Quick reference

To	Do
Silence the buzzer	Press SILENCE.
Silence the hooters	Key switch ON, press SILENCE. Press again to re-sound.
Reset after an alarm	Remove the cause, key switch ON, press RESET.
Raise the alarm	Key switch ON, EVACUATE on Home, OK.
See all fires	Arrow keys on the fire screen, or let the panel show them in turn.
See past events	Events tab.

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EMBER SAFE

STAY ALERT

SmartLoop-4 - 4 Loop Analogue Addressable Fire Alarm Control Panel

Designed and manufactured in India

INFUSIVE INDOTECH PRIVATE LIMITED

Address: _____

Phone: _____ E-mail: _____

Website: _____

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